

# Presentability in the context of higher categories

David Martínez Carpena

Supervisors: Carles Casacuberta & Javier J. Gutiérrez

X Encuentro de Jóvenes Topólogos

This work is supported by the MCIN/ AEI/10.13039/501100011033/ under the I+D+i grant  
PID2020-117971GB-C22

October 20, 2022



X ENCUESTRO DE  
JÓVENES TOPÓLOGOS

# Contents

Higher categories

Higher categories via quasicategories

Presentable higher categories

Presentability and sketches

# Contents

Higher categories

Higher categories via quasicategories

Presentable higher categories

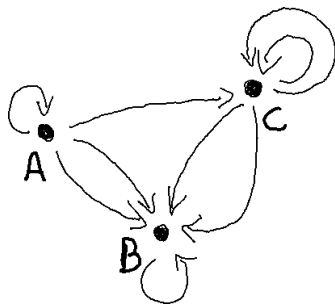
Presentability and sketches

## Infinity categories

In higher category theory, the main objects of study are **higher categories**, which are composed not only of objects and morphisms between objects, but also of  $n$ -morphisms between  $(n - 1)$ -morphisms for all  $n \geq 1$ .

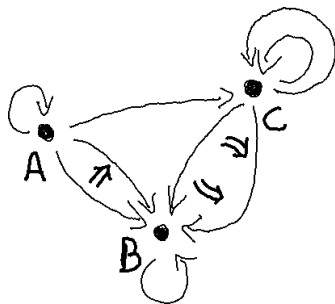
# Infinity categories

In higher category theory, the main objects of study are **higher categories**, which are composed not only of objects and morphisms between objects, but also of  $n$ -morphisms between  $(n - 1)$ -morphisms for all  $n \geq 1$ .



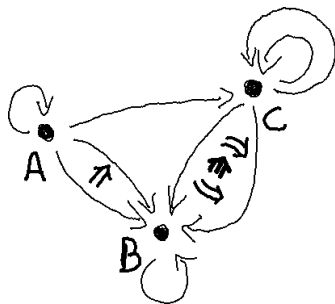
# Infinity categories

In higher category theory, the main objects of study are **higher categories**, which are composed not only of objects and morphisms between objects, but also of  $n$ -morphisms between  $(n - 1)$ -morphisms for all  $n \geq 1$ .



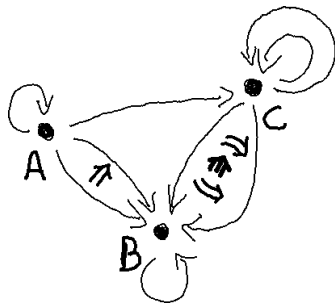
# Infinity categories

In higher category theory, the main objects of study are **higher categories**, which are composed not only of objects and morphisms between objects, but also of  $n$ -morphisms between  $(n - 1)$ -morphisms for all  $n \geq 1$ .



# Infinity categories

In higher category theory, the main objects of study are **higher categories**, which are composed not only of objects and morphisms between objects, but also of  $n$ -morphisms between  $(n - 1)$ -morphisms for all  $n \geq 1$ .



The composition is **weak**:

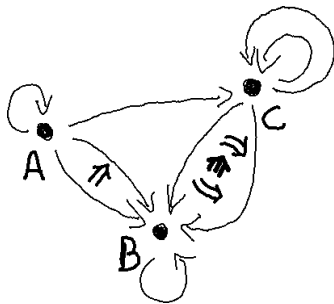
- For any  $n$ -morphisms  $f$ ,  $g$  and  $h$  there is a  $(n + 1)$ -morphism

$$f \circ (g \circ h) \Rightarrow (f \circ g) \circ h$$



# Infinity categories

In higher category theory, the main objects of study are **higher categories**, which are composed not only of objects and morphisms between objects, but also of  $n$ -morphisms between  $(n - 1)$ -morphisms for all  $n \geq 1$ .



The composition is **weak**:

- For any  $n$ -morphisms  $f$ ,  $g$  and  $h$  there is a  $(n + 1)$ -morphism

$$f \circ (g \circ h) \Rightarrow (f \circ g) \circ h$$

- For any  $n$ -morphism  $f$  there is  $(n + 1)$ -morphisms

$$f \circ \text{id} \Rightarrow f \Leftarrow \text{id} \circ f$$

# Infinity categories and Infinity groupoids

A higher category is a  $(\infty, m)$ -category if for any  $n > m$ , the  $n$ -morphisms are invertible up to a  $(n + 1)$ -morphism.

Then, define:

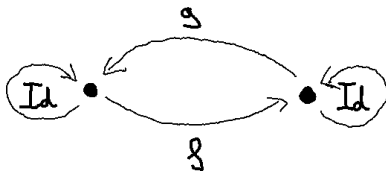
- ▶ An  $\infty$ -**category** as a  $(\infty, 1)$ -category.
- ▶ An  $\infty$ -**groupoid** is a  $(\infty, 0)$ -category.

# Infinity categories and Infinity groupoids

A higher category is a  $(\infty, m)$ -category if for any  $n > m$ , the  $n$ -morphisms are invertible up to a  $(n + 1)$ -morphism.

Then, define:

- ▶ An  $\infty$ -**category** as a  $(\infty, 1)$ -category.
- ▶ An  $\infty$ -**groupoid** is a  $(\infty, 0)$ -category.

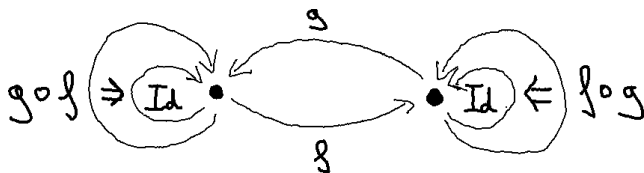


# Infinity categories and Infinity groupoids

A higher category is a  $(\infty, m)$ -category if for any  $n > m$ , the  $n$ -morphisms are invertible up to a  $(n + 1)$ -morphism.

Then, define:

- ▶ An  $\infty$ -**category** as a  $(\infty, 1)$ -category.
- ▶ An  $\infty$ -**groupoid** is a  $(\infty, 0)$ -category.



# Models of higher categories

Historically, there have been many definitions for higher categories, and each one is called a **model**:

- ▶ Globular models [Grothendieck, Batanin, Berger, etc.]
- ▶ Topological (or simplicial) categories [Bergner, Lurie, etc.]
- ▶ Quasicategories [Joyal, Lurie]
- ▶ Complete Segal spaces [Rezk]
- ▶ Segal categories [Segal, Dwyer-Kan-Smith]
- ▶ Relative categories [Dwyer-Kan, Barwick-Kan]

# Contents

Higher categories

Higher categories via quasicategories

Presentable higher categories

Presentability and sketches

# Simplicial sets

The **simplex category**  $\Delta$  is the category with objects the linearly ordered sets  $[n] := \{0, 1, \dots, n\}$  for all  $n \geq 0$ , and morphisms all set functions  $[n] \rightarrow [m]$  which are order-preserving.

# Simplicial sets

The **simplex category**  $\Delta$  is the category with objects the linearly ordered sets  $[n] := \{0, 1, \dots, n\}$  for all  $n \geq 0$ , and morphisms all set functions  $[n] \rightarrow [m]$  which are order-preserving.

A **simplicial set** is a functor  $X : \Delta^{\text{op}} \rightarrow \mathbf{Set}$ . Simplicial sets with natural transformations form a category **sSet**.



# Simplicial sets

The **simplex category**  $\Delta$  is the category with objects the linearly ordered sets  $[n] := \{0, 1, \dots, n\}$  for all  $n \geq 0$ , and morphisms all set functions  $[n] \rightarrow [m]$  which are order-preserving.

A **simplicial set** is a functor  $X : \Delta^{\text{op}} \rightarrow \mathbf{Set}$ . Simplicial sets with natural transformations form a category **sSet**.

For each  $[n]$ , a simplicial set  $X$  has a set denoted  $X[n]$  or  $X_n$ . In addition to those sets,  $X$  is determined by its **faces**  $d_i^n : X_n \rightarrow X_{n-1}$  and **degeneracies**  $s_i^n : X_n \rightarrow X_{n+1}$ , which satisfy the simplicial identities.

## Idea of simplicial sets

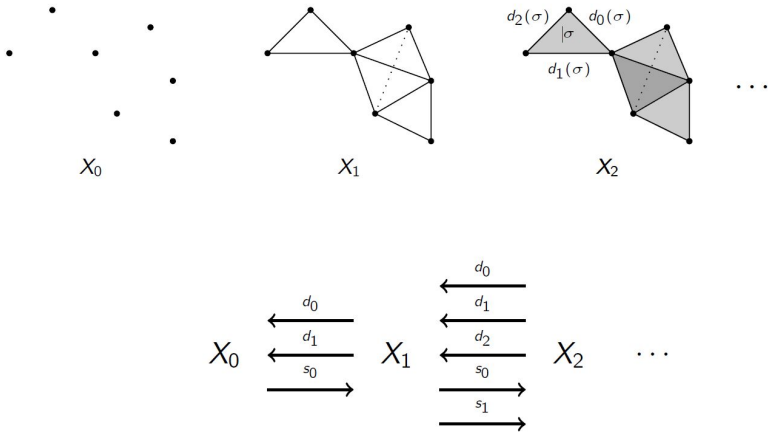


Figure: A simplicial set  $X : \Delta^{\text{op}} \rightarrow \mathbf{Set}$ , from nLab wiki.

## Standard simplices and horns

The **standard  $n$ -simplex** is the simplicial set defined by

$$\Delta^n := \Delta(-, [n]).$$

## Standard simplices and horns

The **standard  $n$ -simplex** is the simplicial set defined by

$$\Delta^n := \Delta(-, [n]).$$

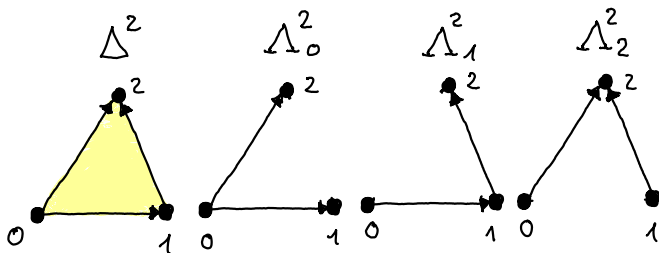
The  **$k$ -th horn**  $\Lambda_k^n$  is the sub-simplicial-set of  $\Delta^n$  obtained from removing the  $k$ -th face.

# Standard simplices and horns

The **standard  $n$ -simplex** is the simplicial set defined by

$$\Delta^n := \Delta(-, [n]).$$

The  $k$ -**th horn**  $\Lambda_k^n$  is the sub-simplicial-set of  $\Delta^n$  obtained from removing the  $k$ -th face.



# Quasi-categories and Kan complexes

$X \in \mathbf{sSet}$  has the  **$k$ -th horn extension property** if for every  $n \in \mathbb{N}$  and every map  $f : \Lambda_k^n \rightarrow X$ , there exists a map  $\tilde{f} : \Delta^n \rightarrow X$  making the following diagram commute:

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{f} & X \\ i \downarrow & \nearrow \tilde{f} & \\ \Delta^n & & \end{array}$$

# Quasi-categories and Kan complexes

$X \in \mathbf{sSet}$  has the  **$k$ -th horn extension property** if for every  $n \in \mathbb{N}$  and every map  $f : \Lambda_k^n \rightarrow X$ , there exists a map  $\tilde{f} : \Delta^n \rightarrow X$  making the following diagram commute:

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{f} & X \\ i \downarrow & \nearrow \tilde{f} & \\ \Delta^n & & \end{array}$$

A simplicial set is:

- ▶ A **quasicategory** if it has the  $k$ -th horn extension property for all  $0 < k < n$ .
- ▶ A **Kan complex** is a simplicial set that has the  $k$ -th horn extension property for all  $0 \leq k \leq n$ .

## Higher structure

There exists a model of higher categories where quasicategories correspond to  $\infty$ -categories, and Kan complexes correspond to  $\infty$ -groupoids.



## Higher structure

There exists a model of higher categories where quasicategories correspond to  $\infty$ -categories, and Kan complexes correspond to  $\infty$ -groupoids.

In a quasicategory  $\mathcal{C}$  the 0-simplices represent the objects, and the  $n$ -simplices represent  $n$ -morphisms.

## Higher structure

There exists a model of higher categories where quasicategories correspond to  $\infty$ -categories, and Kan complexes correspond to  $\infty$ -groupoids.

In a quasicategory  $\mathcal{C}$  the 0-simplices represent the objects, and the  $n$ -simplices represent  $n$ -morphisms.

It can be shown that for any two quasicategories  $\mathcal{C}$  and  $\mathcal{D}$ , there is a quasicategory of  $\infty$ -**functors** between them, defined by:

$$\mathrm{Fun}(\mathcal{C}, \mathcal{D}) := \mathbf{sSet}(\mathcal{C} \times \Delta^\bullet, \mathcal{D})$$

Here, the 0-simplices are functors, and the 1-simplices are natural transformations

# Mapping space

In an abstract  $\infty$ -category, the set of morphisms between two objects is replaced by an  $\infty$ -groupoid of morphisms, which includes the information about the higher structure.

All the higher structure between two objects  $x$  and  $y$  inside a quasicategory  $\mathcal{C}$  can be represented as a Kan complex, usually called the **mapping space**  $\mathrm{Map}_{\mathcal{C}}(x, y)$ , and defined as the following pullback:

$$\begin{array}{ccc} \mathrm{Map}_{\mathcal{C}}(x, y) & \longrightarrow & \mathrm{Fun}(\Delta^1, \mathcal{C}) \\ \downarrow \lrcorner & & \downarrow p \\ \{(x, y)\} & \hookrightarrow & \mathcal{C} \times \mathcal{C} \end{array}$$

# Contents

Higher categories

Higher categories via quasicategories

**Presentable higher categories**

Presentability and sketches

## Limits and colimits

Let  $K$  be any simplicial set,  $y \in \mathcal{C}_0$  an object, and  $F : K \rightarrow \mathcal{C}$  any functor. Given any object  $x \in \mathcal{C}_0$ , the **constant functor** on  $x$   $\underline{x} : K \rightarrow \mathcal{C}$  sends all objects of  $K$  to  $x$ , and all higher morphisms to higher identities.

## Limits and colimits

Let  $K$  be any simplicial set,  $y \in \mathcal{C}_0$  an object, and  $F : K \rightarrow \mathcal{C}$  any functor. Given any object  $x \in \mathcal{C}_0$ , the **constant functor** on  $x$   $\underline{x} : K \rightarrow \mathcal{C}$  sends all objects of  $K$  to  $x$ , and all higher morphisms to higher identities.

A natural transformation  $\alpha : \underline{y} \Rightarrow F$  **exhibits  $y$  as a limit of  $F$**  if  $\alpha$  induces a homotopy equivalence of Kan complexes

$$\mathrm{Map}_{\mathcal{C}}(x, y) \longrightarrow \mathrm{Map}_{\mathrm{Fun}(K, \mathcal{C})}(\underline{x}, F).$$

## Limits and colimits

Let  $K$  be any simplicial set,  $y \in \mathcal{C}_0$  an object, and  $F : K \rightarrow \mathcal{C}$  any functor. Given any object  $x \in \mathcal{C}_0$ , the **constant functor** on  $x$   $\underline{x} : K \rightarrow \mathcal{C}$  sends all objects of  $K$  to  $x$ , and all higher morphisms to higher identities.

A natural transformation  $\alpha : \underline{y} \Rightarrow F$  **exhibits  $y$  as a limit of  $F$**  if  $\alpha$  induces a homotopy equivalence of Kan complexes

$$\mathrm{Map}_{\mathcal{C}}(x, y) \longrightarrow \mathrm{Map}_{\mathrm{Fun}(K, \mathcal{C})}(\underline{x}, F).$$

A natural transformation  $\beta : F \Rightarrow \underline{y}$  **exhibits  $y$  as a colimit of  $F$**  if  $\beta$  induces a homotopy equivalence of Kan complexes

$$\mathrm{Map}_{\mathcal{C}}(y, z) \longrightarrow \mathrm{Map}_{\mathrm{Fun}(K, \mathcal{C})}(F, \underline{z}).$$

# Accessible and presentable quasicategories

A quasicategory  $\mathcal{C}$  is **accessible** if there is a regular cardinal  $\kappa$  such that:

- ▶  $\mathcal{C}$  is locally small.
- ▶  $\mathcal{C}$  admits  $\kappa$ -filtered colimits
- ▶ The full subcategory  $\mathcal{C}_\kappa \subset \mathcal{C}$  of  $\kappa$ -compact objects is essentially small.
- ▶  $\mathcal{C}_\kappa$  generates  $\mathcal{C}$  under small,  $\kappa$ -filtered colimits.



# Accessible and presentable quasicategories

A quasicategory  $\mathcal{C}$  is **accessible** if there is a regular cardinal  $\kappa$  such that:

- ▶  $\mathcal{C}$  is locally small.
- ▶  $\mathcal{C}$  admits  $\kappa$ -filtered colimits
- ▶ The full subcategory  $\mathcal{C}_\kappa \subset \mathcal{C}$  of  $\kappa$ -compact objects is essentially small.
- ▶  $\mathcal{C}_\kappa$  generates  $\mathcal{C}$  under small,  $\kappa$ -filtered colimits.

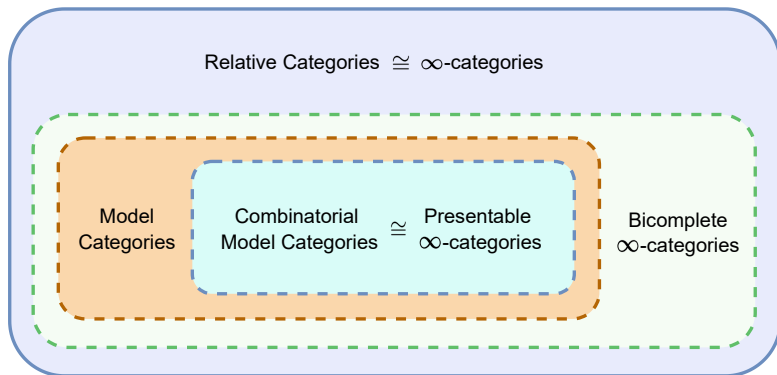
## Definition

A quasicategory  $\mathcal{C}$  is **(locally) presentable** if  $\mathcal{C}$  is accessible and has all small colimits.

# Characterization of presentable quasicategories

## Theorem (Lurie, Pavlov)

*A quasicategory is presentable if, and only if, it is presented by a combinatorial model category.*



# Contents

Higher categories

Higher categories via quasicategories

Presentable higher categories

Presentability and sketches

## Sketches

Recall that a **cone** (resp. **cocone**) of a functor  $F : K \rightarrow C$  at an object  $y$  is a natural transformation  $\alpha : \underline{y} \Rightarrow F$  (resp.  $\beta : F \Rightarrow \underline{y}$ ).

# Sketches

Recall that a **cone** (resp. **cocone**) of a functor  $F : K \rightarrow C$  at an object  $y$  is a natural transformation  $\alpha : \underline{y} \Rightarrow F$  (resp.  $\beta : F \Rightarrow \underline{y}$ ).

## Definition (Ehresmann)

A **sketch**  $\mathcal{S} = (K, \mathfrak{L}, \mathfrak{C})$  is a small category  $K$  equip with a set  $\mathfrak{L}$  of cones and a set  $\mathfrak{C}$  of cocones. If  $\mathfrak{C} = \emptyset$ , we call it a **limit sketch**.

# Sketches

Recall that a **cone** (resp. **cocone**) of a functor  $F : K \rightarrow C$  at an object  $y$  is a natural transformation  $\alpha : \underline{y} \Rightarrow F$  (resp.  $\beta : F \Rightarrow \underline{y}$ ).

## Definition (Ehresmann)

A **sketch**  $\mathcal{S} = (K, \mathfrak{L}, \mathfrak{C})$  is a small category  $K$  equip with a set  $\mathfrak{L}$  of cones and a set  $\mathfrak{C}$  of cocones. If  $\mathfrak{C} = \emptyset$ , we call it a **limit sketch**.

A **model**  $M : K \rightarrow \mathbf{Set}$  of a sketch  $\mathcal{S} = (K, \mathfrak{L}, \mathfrak{C})$  is a functor sending each cone in  $\mathfrak{L}$  to a limit cone, and each cocone in  $\mathfrak{C}$  to a colimit cocone. The category of all models of  $\mathcal{S}$  is denoted  $\text{Mod}(\mathcal{S})$ .

# Representation theorem

## Theorem (Adamek-Rosicky)

*Let  $C$  be a category. The following are equivalent:*

- (i)  $C$  is a locally presentable category.*
- (ii)  $C$  is equivalent to an accessible localization of the category of presheaves  $\mathbf{PSh}(K)$  on a small category  $K$ .*
- (iii)  $C$  is equivalent to the category of models of limit sketches.*

# Representation theorem

## Theorem (Adamek-Rosicky)

*Let  $C$  be a category. The following are equivalent:*

- (i)  $C$  is a locally presentable category.*
- (ii)  $C$  is equivalent to an accessible localization of the category of presheaves  $\mathbf{PSh}(K)$  on a small category  $K$ .*
- (iii)  $C$  is equivalent to the category of models of limit sketches.*

## Theorem (Simpson, Lurie)

*A quasicategory  $\mathcal{C}$  is presentable if, and only if, there exists an accessible localization of the quasicategory of presheaves  $\mathbf{PSh}(\mathcal{K})$  on a small quasicategory  $\mathcal{K}$ .*



# Generalization to higher categories

| Context              | Localization of presheaves | Sketchable     |
|----------------------|----------------------------|----------------|
| Categories           | Adamek-Rosicky             | Adamek-Rosicky |
| Model Categories     | Dugger                     | ?              |
| $\infty$ -categories | Simpson, Lurie             | ?              |

**Table:** Table of the equivalent characterizations seen in the representation theorem but in more general contexts.

# Bibliography



Jacob Lurie. *Higher Topos Theory*. Annals of Mathematics Studies 170. Princeton University Press, 2009.



Jacob Lurie. *Kerodon*. 2022. URL: <https://kerodon.net>.



Charles Rezk. *Introduction to Quasicategories*. June 1, 2022. URL: <https://faculty.math.illinois.edu/~rezk/quasicats.pdf>.



Daniel Dugger. “Combinatorial Model Categories Have Presentations”. In: *Advances in Mathematics* 164.1 (2001).



Jiří Adamek and Jiří Rosicky. *Locally Presentable and Accessible Categories*. Vol. 189. Cambridge University Press, 1994.

# Presentability in the context of higher categories

David Martínez Carpena

Supervisors: Carles Casacuberta & Javier J. Gutiérrez

X Encuentro de Jóvenes Topólogos

This work is supported by the MCIN/ AEI/10.13039/501100011033/ under the I+D+i grant  
PID2020-117971GB-C22

October 20, 2022



X ENCUESTRO DE  
JÓVENES TOPÓLOGOS